

# SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

## Executive Summary

This report compares total site energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. The construction cost data is from 2026 RSMeans Database. Since RSMeans Database is proprietary, users can adopt customized cost dataset when needed.

## Building Information

- Weather File: USA\_NY\_Buffalo-Greater.Buffalo.Intl.AP.725280\_TMY3.epw
- Building Name: SmallOffice
- Building Location: Buffalo, NY
- Total Floor Area: 511 m<sup>2</sup>

## Renovation Details by Scenario

| Scenario   | Renovation Type             | Renovation Details                                                                                                                                                                                                                                                                                                                                                        |
|------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Baseline   | baseline                    | Baseline (no envelope renovation)                                                                                                                                                                                                                                                                                                                                         |
| Scenario 1 | wall + roof + window + door | <p><b>Wall upgrade:</b></p> <ul style="list-style-type: none"><li>• Exterior wall insulation: Blown Fiberglass, target R-20.4</li></ul> <p><b>Roof upgrade:</b></p> <ul style="list-style-type: none"><li>• Roof insulation: Blown Fiberglass, target R-34.5</li></ul> <p><b>Window upgrade:</b></p> <ul style="list-style-type: none"><li>• 2-pane replacement</li></ul> |

|  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  | <ul style="list-style-type: none"><li>Weatherstrip application (skipped: no OperableWindow)</li><li>Glazing film application: anti-graffiti film</li><li>Caulking application: polyurethane</li><li>Window frame replacement: wood window frame</li></ul> <p><b>Door upgrade:</b></p> <ul style="list-style-type: none"><li>Bottom seal: silicone adhesive smoke gasket</li><li>Top/side seal: silicone adhesive smoke gasket</li></ul> |
|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Annual Energy Analysis

| Scenario   | Metric                 | Baseline | Renovation | Delta | Savings % |
|------------|------------------------|----------|------------|-------|-----------|
| Scenario 1 | Total Site Energy (GJ) | 440.31   | 388.03     | 52.28 | 11.87%    |

Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) (\$/yr) Comparison

|            |             |          |
|------------|-------------|----------|
| Baseline   | <div></div> | \$18,286 |
| Scenario 1 | <div></div> | \$17,052 |

■ Baseline ■ Scenario 1

Max saving: **\$-1,234 (-6.75%)**

Total Site Energy Saving (Scenario - Baseline) (GJ/yr) Comparison

|            |             |     |
|------------|-------------|-----|
| Baseline   | <div></div> | 440 |
| Scenario 1 | <div></div> | 388 |

■ Baseline ■ Scenario 1

Max saving: **52 GJ (12%)**

Min Retrofit Construction Cost

**\$69,225**

Scenario 1

Lowest Retrofit Cost Payback Period

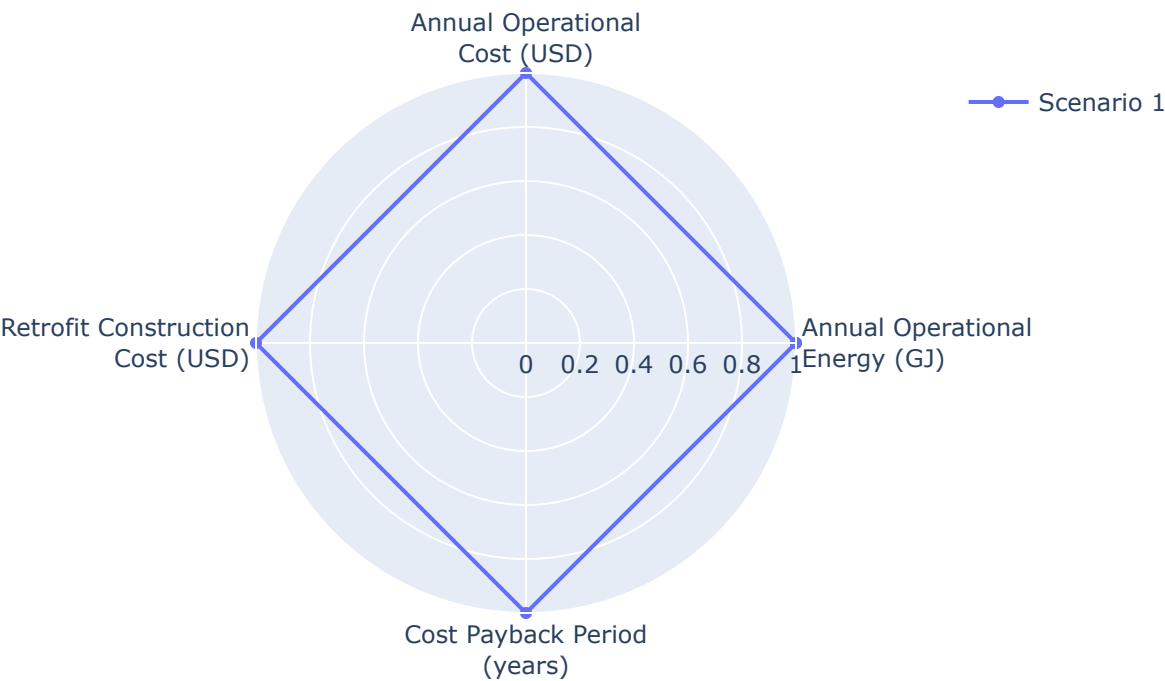
**56 years**

Scenario 1

Comparative Visualizations between Renovation Scenarios

Spider Chart Visualization

Parametric Scenario Comparison (Normalized Spider Chart)



Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) (\$/yr)).

Scenario 1 56 yrs

Material List

Material properties and consumption by renovation scenario. Missing values are shown as N/A.

| Scenario   | Component       | Material name    | Volume (m <sup>3</sup> ) | Area (m <sup>2</sup> ) | Thickness (m) | Length (m) | Thermal Conductivity (W/m·K) | Density (kg/m <sup>3</sup> ) | Product Lifetime (years) |
|------------|-----------------|------------------|--------------------------|------------------------|---------------|------------|------------------------------|------------------------------|--------------------------|
| Scenario 1 | Wall insulation | Blown Fiberglass | 17                       | 282                    | N/A           | N/A        | 0.03                         | 30                           | 30                       |

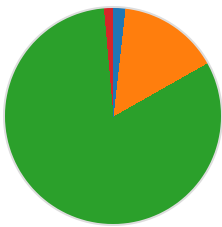
|            |                       |                                |        |      |      |      |      |     |    |
|------------|-----------------------|--------------------------------|--------|------|------|------|------|-----|----|
| Scenario 1 | Roof insulation       | Blown Fiberglass               | 144    | 599  | 0.24 | N/A  | 0.03 | 30  | 30 |
| Scenario 1 | Window frame          | wood window frame              | N/A    | 6.51 | N/A  | N/A  | N/A  | N/A | 15 |
| Scenario 1 | Window caulking       | polyurethane                   | 0.0034 | N/A  | N/A  | N/A  | N/A  | N/A | 10 |
| Scenario 1 | Door bottom sealing   | silicone adhesive smoke gasket | N/A    | N/A  | N/A  | 3.66 | N/A  | N/A | 15 |
| Scenario 1 | Door top/side sealing | silicone adhesive smoke gasket | N/A    | N/A  | N/A  | 16   | N/A  | N/A | 15 |

## Result Summary

Per-scenario breakdown: left chart shows retrofit construction cost contribution by measure, right chart shows total site energy by fuel type (electricity, natural gas, and other). Water is excluded from energy breakdown.

### Scenario 1

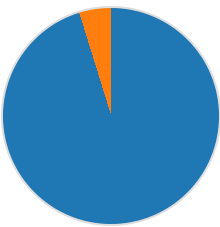
Cost breakdown by retrofit measure



- Wall: \$1,230 (1.8%)
- Roof: \$10,401 (15.0%)
- Window: \$56,623 (81.8%)
- Door: \$971 (1.4%)

Total: \$69,225

Total Site Energy Breakdown (GJ)



- Electricity: 369.41 GJ (95.2%)
- Natural Gas: 18.62 GJ (4.8%)

Total: 388.03 GJ

| Scenario   | Retrofit Construction Cost (USD) | Annual Operational Cost (USD) | Total Site Energy (GJ) |
|------------|----------------------------------|-------------------------------|------------------------|
| Baseline   | \$0.00                           | \$18,286                      | 440.31                 |
| Scenario 1 | \$69,225                         | \$17,052                      | 388.03                 |

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